Type MS · The stored program concept and the parts of the CPU

Answer 1 of 8 · Question 1 part a.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · The stored program concept and the parts of the CPU

Answer 1 of 8 · Question 1 part a.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

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2022 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

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<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
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Type MS · The stored program concept and the parts of the CPU

Answer 1 of 8 · Question 1 part a.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The stored program concept and the parts of the CPU

Answer 2 of 8 · Question 1 part b · 1 mark

2019 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

1(a)	Award **one** from: Input, process, output in this order – **3 marks** Input, process, output in any order – **2 marks** At least one of input, process, output – **1 mark**		3

Question Number	Answer	Additional Guidance	Mark
1(b)	B: Central processing unit **The only correct answer is B** **A** is not correct as this is memory only **C** is not correct as this is memory only **D** is not correct as this is used only for transporting memory addresses		1

Question Number	Answer	Additional Guidance	Mark
1(c)	• Data (1) • Instructions (1)		2

Question Number	Answer	Additional Guidance	Mark
1(d)	Any **two** from: 1. Virtual memory (VM) is used when RAM becomes full (1) (to hold all programs and data). 2. Virtual memory is used as (an extension to) main memory/RAM / works like RAM. (1) 3. Virtual memory is stored/created on (internal) secondary storage/HDD/SSD. (1) 4. Virtual memory is used as temporary storage. (1) 5. Instructions and data not currently being used are transferred from RAM to VM/HDD. (1) 6. When needed again, instructions and data are transferred back to RAM. (1)		2

Type MS · The stored program concept and the parts of the CPU

Answer 2 of 8 · Question 1 part b · 1 mark

2019 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Any **two** from: 1. The disc surface is made out of polycarbonate. (1) 2. A laser (beam) is used. (1) 3. Data is stored as a series of tiny grooves arranged in tracks on the surface of the disc. (1) 4. Bumps/lands/flats/troughs/pits/dips imprinted/burnt on the disc. (1) 5. Bumps and flats (etc.) represent 1 and 0 (bits) / the reflection (of the laser) represents 1 and 0 (bits). (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)	D: System software **The only correct answer is D** **A** is not correct because this category is purely for software that is used for security purposes **B** is not correct because this category is designed for end user tasks rather than for managing the computer itself **C** is not correct because this software is used to translate a high level program into machine code		1

Type MS · The stored program concept and the parts of the CPU

Answer 2 of 8 · Question 1 part b · 1 mark

2019 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The stored program concept and the parts of the CPU

Answer 3 of 8 · Question 1 part c · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

1(a)	Award **one** from: Input, process, output in this order – **3 marks** Input, process, output in any order – **2 marks** At least one of input, process, output – **1 mark**		3

Question Number	Answer	Additional Guidance	Mark
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Type MS · The stored program concept and the parts of the CPU

Answer 3 of 8 · Question 1 part c · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Any **two** from: 1. The disc surface is made out of polycarbonate. (1) 2. A laser (beam) is used. (1) 3. Data is stored as a series of tiny grooves arranged in tracks on the surface of the disc. (1) 4. Bumps/lands/flats/troughs/pits/dips imprinted/burnt on the disc. (1) 5. Bumps and flats (etc.) represent 1 and 0 (bits) / the reflection (of the laser) represents 1 and 0 (bits). (1)		2

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Type MS · The stored program concept and the parts of the CPU

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2019 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The stored program concept and the parts of the CPU

Answer 4 of 8 · Question 4 part d.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part d.i in this mark scheme extract

Number		Guidance	
4(a)	Award **one** mark from: • Predict/model outcomes (1) • Investigate hypotheses (1) • Run experiments that cannot be run / are dangerous in real life (1) • Simulations can be speeded up/slowed down (1)		1

Type MS · The stored program concept and the parts of the CPU

Answer 4 of 8 · Question 4 part d.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **two** marks from: • Simulations (may) use/collect large amounts of data (1) • Simulations (may) involve a large number of calculations (1) • Simulations may need more CPU cycles as the amount of data grows (1) • To be of value, the results must be calculated as quickly as possible (1) • Calculations may grow exponentially / become more complex as the simulation runs/is modified (1) • Simulation may output complex graphics (e.g. virtual world) (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)	<table><tr><th>NOT S</th><th>R AND W</th><th>(NOT S) OR (R AND W)</th></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • Correct values in NOT S column (1) • Correct values in R AND W column (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 1 and/or mark point 2	3

Type MS · The stored program concept and the parts of the CPU

Answer 4 of 8 · Question 4 part d.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part d.i in this mark scheme extract

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• Instructions (and data) are stored in (main) memory (1)
• Instructions (and data) are fetched, decoded, and executed (in a sequence by the CPU) (1)</td><td>Must include storage and execution for 2 marks</td><td>2</td></tr>
<tr><td>4(d)(ii)</td><td>Award two marks from:

• Program counter / PC (1)
• Current instruction (register) / CIR (1)
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Address of instruction carried on address bus (to main memory) (1) contents at that address carried back (to the CPU) on the data bus (1)</td><td></td><td>2</td></tr>
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Type MS · The stored program concept and the parts of the CPU

Answer 4 of 8 · Question 4 part d.i · 2 marks

2021 JUN · Paper 01 · extract 4 of 4 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The stored program concept and the parts of the CPU

Answer 5 of 8 · Question 4 part d.ii · 2 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part d.ii in this mark scheme extract

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Type MS · The stored program concept and the parts of the CPU

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2021 JUN · Paper 01 · extract 2 of 4 · find part d.ii in this mark scheme extract

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Type MS · The stored program concept and the parts of the CPU

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2021 JUN · Paper 01 · extract 4 of 4 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The stored program concept and the parts of the CPU

Answer 6 of 8 · Question 5 part a · 5 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

5(a)	Award **one** mark for each correct component: **Component** / **Letter** Control unit — B Output device — G Registers — C Main memory — H Cache — F Input device — E Arithmetic/logic unit (ALU) — D Clock — A		5

Type MS · The stored program concept and the parts of the CPU

Answer 6 of 8 · Question 5 part a · 5 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark each to a maximum of **two** marks for: • stores a memory address (for the next instruction) (1) • memory address incremented/changed during the fetch process (1)		2
5(b)(ii)	**The only correct answer is B** A *is not correct because* there is no such signal C *is not correct because* as there is no such signal D *is not correct because* as that does not take place during the fetch stage		1
5(b)(iii)	Award **one** mark from: • processes more instructions in the same amount of time (1) • more cycles can be performed in the same amount of time (1) • fetch-execute-decode cycle runs faster (1)		1

Type MS · The stored program concept and the parts of the CPU

Answer 6 of 8 · Question 5 part a · 5 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Award **one** mark from: • a system designed for one/a few specific functions/task(s) (1) • has both hardware and software (1) • has integrated memory (1) • has no or minimal user interface (1) • is power efficient/low power consumption (1) • its functionality cannot be changed/upgraded by users (1) • often use sensors and actuators to interact with the external environment (1) • functions in real time (1)		2
5(c)(ii)	Award up to **two** marks for a linked explanation such as: • Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1) • Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1) • The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1) • Only needs a small bus width (1) because the output signals are serial binary (1)		2
		Total	**13**

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Answer 6 of 8 · Question 5 part a · 5 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

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2022 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

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